

On the Existence and Stability of Recursive Semiotic Fields: A Formalization of the Creative Determinant

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Abstract

We introduce the *Creative Determinant* (CD), a recursive field formalism intended to characterize systems that exhibit coupled coherence–exploration behavior. Mathematically, we work on a compact Riemannian manifold M (the *semiotic manifold*) and study a nonlinear elliptic boundary value problem (BVP) with logistic saturation,

$$-\Delta\Phi = a(x) |\nabla\Phi| + b(x)\Phi - c(x)\Phi^p \quad \text{in } M, \quad \Phi = 0 \quad \text{on } \partial M,$$

where $a(x) = \kappa(x)\gamma(x)\mu(x) \in [0, 1]$ encodes the creative drive (care $\times$ coherence $\times$ contradiction), $b(x)$ is a viability potential admitting a canonical closure $b = \kappa\gamma - \lambda\mu$, and $c(x)\Phi^p$ enforces carrying capacity. We establish existence of *weak coherent configurations* using Schaefer’s fixed-point theorem (Leray–Schauder continuation) with explicit a priori bounds, and prove a nontriviality theorem: when the principal eigenvalue $\lambda_1(-\Delta - b; M) < 0$ (the *viability threshold* is exceeded), the BVP admits a positive solution. Beyond the PDE core, we present an interpretive layer distinguishing weak from strong coherence, formulate conditional geometric statements (a conservation law for a blessing–contradiction 1-form under an equilibrium ansatz; a care-weighted homotopy/energy principle), and propose an operational *Creative Determinant condition* $CD(\alpha, \varepsilon, \delta)$ linking coherence observables to Jacobian volume deformation.

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1 Introduction (interdisciplinary orientation)

Symbolic and subsymbolic approaches to cognition and computation are often treated as fundamentally distinct: symbols as discrete tokens vs. patterns as distributed fields. This paper develops a synthesis in which “symbolic” structure is represented as a continuous field over a manifold of meanings. The resulting framework—the *Creative Determinant* (CD)—formalizes the emergence and maintenance of coherent “presence” through the interaction of four ingredients: (i) gradient structure, (ii) care/adaptivity, (iii) coherence/consistency, and (iv) contradiction/curvature.

The manuscript is structured intentionally for an interdisciplinary audience while maintaining a rigorous mathematical spine:

- **Mathematical core (Sections 2–3).** A nonlinear elliptic PDE with logistic saturation on a compact Riemannian manifold, with a complete existence proof using Schaefer/Leray–Schauder machinery, and a nontriviality theorem using eigenvalue thresholds and sub-/super-solution barriers.
- **Interpretive/structural layer (Section 4).** A careful distinction between weak (PDE) and strong (pointwise) coherence; conditional geometric statements framed as explicit ansätze rather than theorems about the PDE.
- **Operational layer (Section 5).** A testable dynamical-systems condition and empirical falsification protocols, stated as proposals rather than proved consequences.

Main mathematical contributions

- (i) **Existence (Theorem 3.11):** Under mild coefficient assumptions, the autonomous viability model with saturation admits at least one weak coherent configuration.
- (ii) **Nontriviality (Theorem 3.15):** When the viability threshold is exceeded ($\lambda_1(-\Delta - b; M) < 0$), there exists a positive solution—coherent presence can be self-maintained.
- (iii) **Canonical viability closure (Definition 3.3):** The viability potential $b = \kappa\gamma - \lambda\mu$ provides a principled link between the PDE coefficient and the CD characteristic fields.

Main mathematical object and scope

The primary analytic object is the Dirichlet BVP with logistic saturation

$$\begin{cases} -\Delta\Phi = a(x) |\nabla\Phi| + b(x) \Phi - c(x) \Phi^p & \text{in } M, \\ \Phi = 0 & \text{on } \partial M, \end{cases} \quad (1)$$

where $a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1]$, $b \in C^{0,\alpha}(M)$, $c(x) \geq c_0 > 0$, and $p > 1$. Solutions of (1) are called *weak coherent configurations*. The pointwise CD constraint $\Phi = |\nabla\Phi| \kappa\gamma\mu$ is treated as a constitutive idealization that motivates (1), but is not asserted equivalent; clarifying their relationship is an explicit open problem.

2 Mathematical setup

2.1 Semiotic manifold and function spaces

Definition 2.1 (Semiotic manifold). Let (M, g) be a compact, connected, smooth Riemannian manifold of dimension $n \geq 1$, possibly with smooth boundary ∂M . We denote by ∇ the Riemannian gradient and by Δ the Laplace–Beltrami operator.

Remark 2.2 (Terminology: “semiotic”). The term *semiotic manifold* reflects the framework’s roots in Peircean semiotics [1], where meaning emerges through sign relations. The manifold M is interpreted as a space of possible meanings or interpretations.

We use standard Hölder spaces $C^{k,\alpha}(M)$ for $k \in \mathbb{N}$, $\alpha \in (0, 1)$, with the usual norms; see [2, 3]. We write

$$X := C^{1,\alpha}(M) \cap C^0(\overline{M}), \quad X_0 := \{\psi \in X : \psi|_{\partial M} = 0\},$$

and we interpret $|\nabla\psi|$ pointwise using the Riemannian norm induced by g .

2.2 Core scalar fields and the coefficient a

Definition 2.3 (Characteristic fields). Let $\kappa, \gamma, \mu \in C^{0,\alpha}(M)$ be scalar fields with values in $[0, 1]$. Define

$$a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1].$$

Remark 2.4 (Primary boundary condition). Dirichlet boundary conditions are taken as primary for analytic clarity and interpretive meaning: $\Phi = 0$ on ∂M represents boundary contexts where sustained presence/coherence is absent. Neumann conditions are discussed only as a secondary variant.

3 Main PDE Results

3.1 Primary model: autonomous viability with logistic saturation (V1')

Let

$$a(x) := \kappa(x)\gamma(x)\mu(x) \in [0, 1]$$

be the CD modulation coefficient. We introduce a *viability potential* b and a *saturation (carrying capacity) field* c to obtain a mathematically nondegenerate autonomous equilibrium model.

Definition 3.1 (Autonomous viability model with saturation (V1')). Fix $p > 1$. Given coefficients

$$a \in C^{0,\alpha}(M), \quad 0 \leq a \leq 1; \quad b \in C^{0,\alpha}(M); \quad c \in C^{0,\alpha}(M), \quad c(x) \geq c_0 > 0,$$

we consider the Dirichlet problem

$$\begin{cases} -\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi - c(x)\Phi^p & \text{in } M, \\ \Phi = 0 & \text{on } \partial M. \end{cases} \quad (2)$$

A solution $\Phi \in C^{2,\alpha}(M)$ of (2) is called a *weak coherent configuration* for the autonomous CD model.

Remark 3.2 (Interpretive glossary for (2)). In (2):

- $a(x) = \kappa\gamma\mu$ is the *creative drive* term: gradient activity contributes to presence only where care, coherence, and contradiction jointly support it.
- $b(x) > 0$ is *viability gain* / autopoietic support: presence amplifies where sustainable.
- $b(x) < 0$ is a *dissolution zone*: presence decays locally.
- $c(x)\Phi^p$ enforces *carrying capacity* / resource limitation: it prevents unbounded growth and encodes finite-resource autopoiesis.

Definition 3.3 (Canonical viability closure). Under the *canonical viability closure*, the viability potential is defined by

$$b(x) := \kappa(x)\gamma(x) - \lambda\mu(x), \quad (3)$$

where $\lambda > 0$ is a contradiction-weighting parameter.

Remark 3.4 (Role of λ). The parameter λ controls the effective cost of contradiction: small λ makes contradiction cheap (viability easier), while large λ makes contradiction costly (viability requires stronger $\kappa\gamma$). This supports future empirical work in which λ is estimated from observed dynamics.

94 *Remark 3.5* (Limiting case: the homogeneous model (V1)). If $c \equiv 0$, (2) reduces to the positively
 95 homogeneous equation

$$-\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi, \quad \Phi|_{\partial M} = 0,$$

96 which is an idealized “no-saturation” limit. The saturation term is mathematically and inter-
 97 pretively important: it breaks homogeneity and enables barrier constructions yielding nontrivial
 98 equilibria.

99 3.2 Existence of weak coherent configurations (Schaefer/Leray–Schauder)

100 We now prove existence of solutions to (2). The proof follows the same compactness spine as the
 101 original Schauder approach, but uses Schaefer’s fixed-point theorem [4] (a variant of the Leray–
 102 Schauder continuation principle [5]) to avoid invariant-ball vulnerabilities.

103 Operator formulation

104 Let

$$X := \{\psi \in C^{1,\alpha}(M) : \psi|_{\partial M} = 0\}, \quad Y := C^{0,\alpha}(M).$$

105 For $\psi \in X$, define

$$F_\psi(x) := a(x)|\nabla\psi(x)| + b(x)\psi(x) - c(x)(\psi_+(x))^p, \quad \psi_+ := \max\{\psi, 0\}.$$

106 Let $S : Y \rightarrow C^{2,\alpha}(M) \cap X$ denote the Dirichlet solution operator for $-\Delta u = f$ in M , $u|_{\partial M} = 0$.
 107 Define the compact operator

$$T : X \rightarrow X, \quad T(\psi) := S(F_\psi).$$

108 **Lemma 3.6** (Compactness). *The operator $T : X \rightarrow X$ is continuous and compact.*

109 *Proof.* The map $\psi \mapsto |\nabla\psi|$ is continuous $C^{1,\alpha} \rightarrow C^{0,\alpha}$, multiplication by $a, b, c \in C^{0,\alpha}$ is continuous
 110 on $C^{0,\alpha}$, and $\psi \mapsto (\psi_+)^p$ is continuous $C^{0,\alpha} \rightarrow C^{0,\alpha}$ for $p > 1$. Thus $\psi \mapsto F_\psi$ is continuous $X \rightarrow Y$.
 111 The Dirichlet solution operator $S : Y \rightarrow C^{2,\alpha} \cap X$ is continuous and maps bounded sets in Y into
 112 bounded sets in $C^{2,\alpha}$; see [2, Ch. 6]. Since the embedding $C^{2,\alpha}(M) \hookrightarrow C^{1,\alpha}(M)$ is compact on
 113 compact M (Arzelà–Ascoli), the composition T is compact $X \rightarrow X$. $\square$

114 A priori bounds for Schaefer

115 We use Schaefer’s theorem: it suffices to show that the set

$$\mathcal{S} := \{u \in X : u = \tau T(u) \text{ for some } \tau \in [0, 1]\}$$

116 is bounded in X .

117 **Assumption 3.7** (Standard coercivity sign condition on b). There exists $B \geq 0$ such that $b(x) \leq B$
 118 for all $x \in M$.

119 *Remark 3.8.* Assumption 3.7 is mild (automatic if $b \in C^{0,\alpha}$) and is used only to control the linear
 120 reaction term from above in the a priori estimates.

121 **Lemma 3.9** (Uniform L^∞ bound for Schaefer set). *Assume 3.7. Then there exists $K > 0$, depend-*
 122 *ing only on B, c_0, p , such that every $u \in \mathcal{S}$ satisfies*

$$\|u\|_{L^\infty(M)} \leq K.$$

123 *Proof.* Let $u \in \mathcal{S}$. Then $u = \tau T(u)$ for some $\tau \in [0, 1]$, hence $u \in C^{2,\alpha}(M) \cap X$ solves

$$-\Delta u = \tau \left(a|\nabla u| + b u - c(u_+)^p \right), \quad u|_{\partial M} = 0. \quad (4)$$

124 Let $x_0 \in \overline{M}$ be a point where u attains its maximum. Since $u = 0$ on ∂M , either $\max u \leq 0$ or the
 125 maximum is achieved in the interior and is positive. Assume $\max_M u = u(x_0) > 0$. Then $x_0 \in M^\circ$,
 126 $\nabla u(x_0) = 0$, and $\Delta u(x_0) \leq 0$, so $-\Delta u(x_0) \geq 0$. Evaluating (4) at x_0 and using $\nabla u(x_0) = 0$ gives

$$0 \leq -\Delta u(x_0) = \tau \left(b(x_0) u(x_0) - c(x_0) u(x_0)^p \right) \leq \tau \left(B u(x_0) - c_0 u(x_0)^p \right).$$

127 Since $u(x_0) > 0$ and $\tau \geq 0$, this implies

$$B u(x_0) - c_0 u(x_0)^p \geq 0 \quad \implies \quad u(x_0)^{p-1} \leq \frac{B}{c_0}.$$

128 Hence

$$\max_M u \leq \left(\frac{B}{c_0} \right)^{1/(p-1)} =: K.$$

129 If $\max_M u \leq 0$ then the same bound holds trivially. $\square$

130 **Lemma 3.10** (Uniform $C^{1,\alpha}$ bound for Schaefer set). *Assume 3.7. Then $\mathcal{S}$ is bounded in $X = C^{1,\alpha}$.*

131 *Proof.* Let $u \in \mathcal{S}$ solve (4). By Lemma 3.9, $\|u\|_{L^\infty} \leq K$. We estimate the forcing in $C^{0,\alpha}$:

$$f_u := \tau \left(a|\nabla u| + b u - c(u_+)^p \right).$$

132 The terms $b u$ and $c(u_+)^p$ are bounded in $C^{0,\alpha}$ in terms of $\|b\|_{C^{0,\alpha}}$, $\|c\|_{C^{0,\alpha}}$ and $\|u\|_{C^{0,\alpha}} \leq K$. The
 133 remaining term $a|\nabla u|$ is bounded in C^0 by $\|\nabla u\|_{C^0}$. Using standard elliptic regularity in Hölder
 134 spaces (Schauder estimates; [3, Ch. 6]) in the form

$$\|u\|_{C^{2,\alpha}} \leq C \left(\|f_u\|_{C^{0,\alpha}} + \|u\|_{C^0} \right),$$

135 and the interpolation inequality $\|\nabla u\|_{C^0} \leq C_* \|u\|_{C^{2,\alpha}}$, one can absorb the gradient term and obtain
 136 a bound

$$\|u\|_{C^{2,\alpha}} \leq C' \left(1 + \|u\|_{C^0} \right) \leq C''.$$

137 Consequently $\|u\|_{C^{1,\alpha}}$ is uniformly bounded. $\square$

138 **Theorem 3.11** (Existence of weak coherent configurations for (2)). *Assume the coefficient hy-*
 139 *potheses of Definition 3.1 and the upper bound Assumption 3.7. Then the Dirichlet problem (2)*
 140 *admits at least one solution $\Phi \in C^{2,\alpha}(M)$. Moreover, there exists a nonnegative solution $\Phi \geq 0$.*

141 *Proof.* By Lemma 3.6, $T : X \rightarrow X$ is continuous and compact. By Lemmas 3.9–3.10, the Schaefer
 142 set $\mathcal{S}$ is bounded in X . Therefore, by Schaefer's fixed-point theorem [4, 6], T has a fixed point
 143 $\Phi \in X$, which solves (2). Elliptic regularity upgrades Φ to $C^{2,\alpha}(M)$.

144 To obtain a nonnegative solution, note that the operator uses ψ_+ in the saturation term and
 145 the PDE preserves the ordering under standard maximum principle arguments. $\square$

3.3 Nontriviality (Theorem 3.1b): viability threshold and barrier construction

We now prove a genuine nontriviality theorem using the principal eigenvalue of the linear operator $-\Delta - b$.

Definition 3.12 (Principal eigenvalue and viability threshold). Let $\lambda_1(-\Delta - b; M)$ denote the principal Dirichlet eigenvalue of the operator $\mathcal{L}_b u := -\Delta u - b(x)u$ on M . We say the *viability threshold is exceeded* if

$$\lambda_1(-\Delta - b; M) < 0.$$

Remark 3.13 (Spectral theory). The existence of a principal eigenvalue with positive eigenfunction for uniformly elliptic operators follows from the Krein–Rutman theorem [7]; see also [3, Ch. 2] for the PDE context.

Remark 3.14 (Interpretation of the viability threshold). Under the canonical closure $b = \kappa\gamma - \lambda\mu$ (Definition 3.3), the spectral condition $\lambda_1(-\Delta - b; M) < 0$ becomes a precise “*viability exceeds dissipation*” criterion: integrated care–coherence support dominates contradiction load strongly enough to overcome boundary dissipation imposed by $\Phi|_{\partial M} = 0$.

Theorem 3.15 (Existence of nontrivial weak coherent configurations). *Assume the hypotheses of Theorem 3.11. Suppose additionally that*

$$\lambda_1(-\Delta - b; M) < 0.$$

Then (2) admits a nontrivial nonnegative solution $\Phi \in C^{2,\alpha}(M)$, $\Phi \not\equiv 0$. In fact, $\Phi > 0$ in M° .

Proof. Let $\phi_1 \in C^{2,\alpha}(M) \cap C^0(\overline{M})$ be the positive principal eigenfunction of $\mathcal{L}_b$, normalized so that $\|\phi_1\|_{L^\infty(M)} = 1$. Thus

$$-\Delta\phi_1 - b(x)\phi_1 = \lambda_1\phi_1, \quad \phi_1|_{\partial M} = 0,$$

with $\lambda_1 < 0$. Write $\beta := -\lambda_1 > 0$; then

$$-\Delta\phi_1 = b(x)\phi_1 - \beta\phi_1. \tag{5}$$

Step 1: sub-solution. Define $\underline{\Phi} := \varepsilon\phi_1$ with $\varepsilon > 0$ to be chosen. Using (5),

$$-\Delta\underline{\Phi} = \varepsilon b\phi_1 - \varepsilon\beta\phi_1.$$

On the other hand,

$$a|\nabla\underline{\Phi}| + b\underline{\Phi} - c(\underline{\Phi})^p = \varepsilon a|\nabla\phi_1| + \varepsilon b\phi_1 - c\varepsilon^p\phi_1^p.$$

To show $\underline{\Phi}$ is a sub-solution, we need

$$\varepsilon b\phi_1 - \varepsilon\beta\phi_1 \leq \varepsilon a|\nabla\phi_1| + \varepsilon b\phi_1 - c\varepsilon^p\phi_1^p.$$

Cancel the common term $\varepsilon b\phi_1$ and divide by $\varepsilon > 0$:

$$-\beta\phi_1 \leq a|\nabla\phi_1| - c\varepsilon^{p-1}\phi_1^p.$$

Choose $\varepsilon > 0$ such that

$$\varepsilon^{p-1} \leq \frac{\beta}{2\|c\|_{L^\infty(M)}}, \tag{6}$$

so that $c\varepsilon^{p-1}\phi_1^p \leq (\beta/2)\phi_1$ (since $\phi_1 \leq 1$). Then

$$a|\nabla\phi_1| - c\varepsilon^{p-1}\phi_1^p \geq -\frac{\beta}{2}\phi_1,$$

and hence

$$-\beta\phi_1 \leq -\frac{\beta}{2}\phi_1 \leq a|\nabla\phi_1| - c\varepsilon^{p-1}\phi_1^p,$$

since $a|\nabla\phi_1| \geq 0$. Thus $\underline{\Phi}$ is a sub-solution. Also $\underline{\Phi} = 0$ on ∂M .

Step 2: Dirichlet-compatible super-solution. Let $\psi \in C^{2,\alpha}(M) \cap C^0(\overline{M})$ be the unique solution of

$$-\Delta\psi = 1 \text{ in } M, \quad \psi = 0 \text{ on } \partial M.$$

By the maximum principle, $\psi > 0$ in M° . Define $\overline{\Phi} := K\psi$ with $K > 0$ to be chosen. Then

$$-\Delta\overline{\Phi} = K.$$

We require $\overline{\Phi}$ to satisfy

$$K \geq a|K\nabla\psi| + b(K\psi) - c(K\psi)^p = Ka|\nabla\psi| + Kb\psi - cK^p\psi^p.$$

Divide by $K > 0$:

$$1 \geq a|\nabla\psi| + b\psi - cK^{p-1}\psi^p.$$

Since ψ and $\nabla\psi$ are continuous on $\overline{M}$, the terms $a|\nabla\psi| + b\psi$ are bounded above by some constant C_ψ . Choose K large enough that for all $x \in M$,

$$c_0K^{p-1}\psi(x)^p \geq a(x)|\nabla\psi(x)| + b(x)\psi(x) - 1,$$

which is possible because the left-hand side grows like K^{p-1} and the right-hand side is K -independent. Then $\overline{\Phi} = K\psi$ is a super-solution and $\overline{\Phi} = 0$ on ∂M .

Step 3: ordering and conclusion. By construction, $\underline{\Phi} = \varepsilon\phi_1 \geq 0$ and $\overline{\Phi} = K\psi \geq 0$. Choose K further so that $\underline{\Phi} \leq \overline{\Phi}$ in M (possible since $\underline{\Phi}$ is bounded and $\overline{\Phi}$ scales with K). Then by the classical sub-/super-solution theorem for Dirichlet problems [2, Thm. 10.2] (see also [8] for a comprehensive treatment), there exists a solution Φ of (2) such that

$$\underline{\Phi} \leq \Phi \leq \overline{\Phi} \text{ in } M.$$

Since $\underline{\Phi} \not\equiv 0$ and $\underline{\Phi} > 0$ on M° , it follows that $\Phi \not\equiv 0$ and $\Phi \geq 0$. Finally, applying the strong maximum principle to (2) yields $\Phi > 0$ in M° . $\square$

3.4 Master equation and the driven/coupled case

Remark 3.16 (Master equation and specializations). For generality one may consider the master form

$$-\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi - c(x)\Phi^p + f(x), \quad \Phi|_{\partial M} = 0,$$

where f encodes exogenous significance / environmental input.

- **Autonomous (closed) case:** $f \equiv 0$, yielding (2).
- **Driven (coupled) case:** $b \equiv 0$, $c \equiv 0$, yielding a forced, interaction-sustained presence model.

4 Interpretive/structural layer

This section is conditional by design: it collects definitions and geometric statements that connect the PDE layer to the broader Creative Determinant framework.

4.1 Weak vs. strong coherence (logical status)

Definition 4.1 (Pointwise CD constraint (strong coherence)). A *strong coherent configuration* is a field Φ satisfying

$$\Phi(x) = |\nabla \Phi(x)| \kappa(x) \gamma(x) \mu(x) \quad \text{for all } x \in M.$$

Definition 4.2 (Weak coherence). A *weak coherent configuration* is a solution Φ of an associated PDE model (e.g. (2)) with stated boundary conditions.

Remark 4.3 (Relationship is open). Strong coherence implies weak coherence only for PDE models derived from the pointwise constraint under additional assumptions. In the present paper, the PDE is treated as an analytically tractable proxy; no equivalence is claimed.

4.2 Conditional conservation of blessing–contradiction coupling

Definition 4.4 (Blessing–contradiction 1-form). Let $\Phi, \mu \in C^1(M)$. Define the 1-form

$$\omega := \Phi d\mu - \mu d\Phi.$$

Proposition 4.5 (Conditional conservation under equilibrium ansatz). *Let $U \subseteq M$ be open, and assume an equilibrium ansatz on U : there exists a C^1 function $g : [0, 1] \rightarrow \mathbb{R}$ such that*

$$\Phi(x) = g(\mu(x)) \quad \text{for all } x \in U.$$

Then ω is closed on U , i.e. $d\omega = 0$ on U . Consequently, for any two homotopic loops $\Gamma_1, \Gamma_2 \subset U$,

$$\int_{\Gamma_1} \omega = \int_{\Gamma_2} \omega.$$

Proof. Compute

$$d\omega = d(\Phi d\mu) - d(\mu d\Phi) = d\Phi \wedge d\mu - d\mu \wedge d\Phi = 2 d\Phi \wedge d\mu.$$

Under the ansatz $\Phi = g(\mu)$, we have $d\Phi = g'(\mu) d\mu$, hence $d\Phi \wedge d\mu = 0$ and thus $d\omega = 0$ on U . Homotopy invariance follows by Stokes' theorem. $\square$

Remark 4.6 (Interpretation). The ansatz $\Phi = g(\mu)$ expresses a local functional dependence of presence on contradiction, i.e. a phase where the presence field is slaved to contradiction level sets. This is a strong idealization and is stated explicitly as such.

4.3 Care-weighted homotopy/energy principle (geometric interpretation)

Definition 4.7 (Care-weighted metric). On the open set $\{\kappa > 0\} \subseteq M$, define the conformally rescaled metric

$$g_\kappa := \kappa^{-1} g.$$

Definition 4.8 (Care-weighted homotopy energy). Let $H : S^1 \times [0, 1] \rightarrow M$ be a homotopy such that $H(S^1 \times [0, 1]) \subset \{\kappa > 0\}$. Define the energy

$$E_\kappa[H] := \int_0^1 \int_{S^1} \left\| \frac{\partial H}{\partial t} \right\|_{g_\kappa}^2 ds dt.$$

Proposition 4.9 (Care as a sufficient obstruction to resolution). *Let $\gamma_0 : S^1 \rightarrow M$ be a loop. If every homotopy contracting γ_0 to a point must intersect the zero-care set $\{\kappa = 0\}$, then there is no contracting homotopy H with finite $E_\kappa[H]$.*

Proof. On $\{\kappa = 0\}$, the metric $g_\kappa = \kappa^{-1}g$ is undefined (formally infinite). Any homotopy forced to pass through $\{\kappa = 0\}$ cannot be assigned finite g_κ -energy. $\square$

Remark 4.10 (Status). This is a geometric sufficient obstruction statement, not an “iff” characterization: we do not claim that finite-energy homotopy exists whenever a loop is null-homotopic in $\{\kappa > 0\}$, since that requires additional geometric hypotheses.

4.4 Necessity statements: contradiction, care, and coherence

Proposition 4.11 (Local decoupling when contradiction vanishes). *Let Φ be a (weak) coherent configuration solving the autonomous saturated CD model*

$$-\Delta\Phi = a(x)|\nabla\Phi| + b(x)\Phi - c(x)\Phi^p \quad \text{in } M, \quad \Phi|_{\partial M} = 0,$$

with $a(x) = \kappa(x)\gamma(x)\mu(x)$. If $\mu \equiv 0$ on an open set $U \subseteq M$, then $a \equiv 0$ on U and the equation restricts on U to

$$-\Delta\Phi = b(x)\Phi - c(x)\Phi^p \quad \text{in } U.$$

In particular, the CD gradient-coherence coupling term $a|\nabla\Phi|$ vanishes identically on U .

Remark 4.12 (Interpretation: sterile stability). Solutions may persist in regions with $\mu \equiv 0$ and may be stable under the resulting reaction-diffusion balance, but they exhibit no *CD-type creative coupling* on that region, since $a|\nabla\Phi|$ is absent there.

Proposition 4.13 (Care/coherence collapse under the canonical closure). *Assume the canonical closure (Definition 3.3) and $c(x) \geq c_0 > 0$.*

1. *If $\kappa \equiv 0$ on M , then $b = -\lambda\mu \leq 0$ on M .*

2. *If $\gamma \equiv 0$ on M , then $b = -\lambda\mu \leq 0$ on M .*

In either case, the saturated Dirichlet problem admits no nontrivial nonnegative solution: the only solution with $\Phi \geq 0$ is $\Phi \equiv 0$.

Proof. In both cases $b \leq 0$ on M . Suppose $\Phi \geq 0$ is a classical solution and $\Phi \not\equiv 0$. Then Φ attains a positive maximum at some interior point $x_0 \in M^\circ$, where $\nabla\Phi(x_0) = 0$ and $-\Delta\Phi(x_0) \geq 0$. Evaluating the PDE at x_0 gives

$$0 \leq -\Delta\Phi(x_0) = b(x_0)\Phi(x_0) - c(x_0)\Phi(x_0)^p \leq -c_0\Phi(x_0)^p < 0,$$

a contradiction. Hence no such Φ exists. $\square$

Remark 4.14 (Interpretation). Under the canonical closure, vanishing care or coherence forces the viability potential nonpositive ($b \leq 0$), so the saturated dynamics cannot sustain a positive presence field under Dirichlet boundary dissipation.

Proposition 4.15 (Minimal necessity for CD-type coupling). *If a solution Φ exhibits CD-type gradient-coherence coupling on a set of positive measure (i.e. $a(x)|\nabla\Phi(x)| > 0$ on a set of positive measure), then necessarily*

$$\kappa\gamma\mu \neq 0, \quad \text{and in particular} \quad \mu \neq 0, \quad \kappa \neq 0, \quad \gamma \neq 0.$$

Remark 4.16 (The triple conjunction: meaning emerges in tension held with care). The preceding propositions establish a structural constraint on *CD-type creative coupling*. If a weak coherent configuration exhibits active gradient–coherence coupling on a non-negligible region (i.e. $a(x)|\nabla\Phi(x)| > 0$ on a set of positive measure), then necessarily the creative drive coefficient

$$a(x) = \kappa(x)\gamma(x)\mu(x)$$

does not vanish identically. Equivalently, none of the characteristic fields κ, γ, μ can vanish identically, and the product $\kappa\gamma\mu$ must be positive on a region where $|\nabla\Phi|$ is nonzero.

This “triple conjunction” formalizes the principle that *meaning arises between held contradictions*:

- **Contradiction** (μ) provides generative tension—the “difference that makes a difference,” driving exploration along gradients of the semiotic manifold when coupling is active.
- **Coherence** (γ) provides structural integrity—the container that prevents fragmentation under tension, analogous to Husserl’s synthesis in temporal consciousness.
- **Care** (κ) provides holding capacity—the energetic investment that sustains the system through contradiction, preventing collapse into a purely dissipative regime.

The coefficient $a = \kappa\gamma\mu$ is a *product, not a sum*: the vanishing of any single factor annihilates the CD coupling term $a|\nabla\Phi|$ wherever that factor vanishes. In particular:

1. High contradiction with low care (μ large, κ small) places viability under strain; under the canonical closure, $\kappa \equiv 0$ forces $b \leq 0$ and the saturated Dirichlet model admits no nontrivial *nonnegative* solutions (Proposition 4.13).
2. High care and coherence but zero contradiction ($\kappa\gamma$ large, $\mu = 0$) may support stable reaction–diffusion structure, but it exhibits no CD-type creative coupling on that region (Proposition 4.11).
3. Contradiction and care without coherence ($\mu\kappa$ present, $\gamma = 0$) again yields $b \leq 0$ under the canonical closure and thus no nontrivial nonnegative presence can be sustained (Proposition 4.13).

Under the canonical viability closure $b = \kappa\gamma - \lambda\mu$ (Definition 3.3), the parameter $\lambda > 0$ controls contradiction cost: larger λ models systems where sustained tension is expensive (e.g. resource scarcity, accumulated strain), requiring proportionally greater $\kappa\gamma$ to remain viable. The viability criterion can be expressed spectrally via the principal Dirichlet eigenvalue:

$$\lambda_1(-\Delta - b; M) = \lambda_1(-\Delta - \kappa\gamma + \lambda\mu; M).$$

Heuristically, the sign of $\lambda_1(-\Delta - b; M)$ functions as a phase boundary:

- **Below threshold** ($\lambda_1(-\Delta - b; M) > 0$): holding capacity is insufficient; in particular when $b \leq 0$ everywhere, no nontrivial nonnegative solutions exist (Proposition 4.13).
- **Above threshold** ($\lambda_1(-\Delta - b; M) < 0$): holding capacity exceeds contradiction cost; nontrivial coherent presence can emerge in the saturated model (cf. the nontriviality theorem in Section 3).

Empirical predictions. Interventions aimed at enhancing coherence (therapeutic integration, organizational alignment, relationship repair) are predicted to fail if they (i) eliminate contradiction entirely (sterile stability), (ii) proceed without sufficient care/investment (dissolution), or (iii) lack coherence mechanisms (fragmentation). Measurement protocols should therefore track κ , γ , and μ concurrently, with attention to:

1. the product $\kappa\gamma\mu$ as a leading indicator of coupling capacity,
2. the balance $\kappa\gamma - \lambda\mu$ (or its spectral proxy $\lambda_1(-\Delta - b)$) as a proximity-to-threshold metric,
3. temporal changes in each factor as early warning signals of approaching viability crises.

5 Operational layer

This section proposes testable conditions connecting the CD formalism to measurable dynamical systems.

5.1 The Creative Determinant condition

Definition 5.1 (Creative Determinant condition $CD(\alpha, \varepsilon, \delta)$). Let $F : X \rightarrow X$ be a C^1 dynamical system on a smooth manifold X , and let $\varphi : X \rightarrow \mathbb{R}_{>0}$ be a pre-specified coherence observable. We say (F, φ) satisfies $CD(\alpha, \varepsilon, \delta)$ if

$$\text{Corr}[\log \varphi(F(x)), \alpha \log |\det \nabla F(x)|] \geq 1 - \varepsilon \quad \text{and} \quad \text{Var}[\log |\det \nabla F(x)|] > \delta,$$

where correlation/variance are computed empirically along trajectories or ensembles.

Remark 5.2 (Connection to Lyapunov theory). By Oseledets' multiplicative ergodic theorem [9], time averages of $\log |\det \nabla F|$ relate to sums of Lyapunov exponents [10, 11]. The CD condition is intended to detect regimes where a coherence observable tracks cumulative phase-space volume deformation. When the CD condition holds with $\alpha \approx 1$, the correlation suggests the system may exhibit fractal attractor structure, with dimension estimable via the Kaplan–Yorke formula [12].

5.2 Falsifiability protocols (empirical proposals)

We list falsifiability criteria as operational proposals; they are not derived from the PDE theorem.

Definition 5.3 (Framework falsification (empirical)). The CD framework is empirically falsified if any of the following are demonstrated:

- (F1) Coherence without coupling: stable high coherence with near-zero (or negative) correlation between coherence observables and Jacobian volume dynamics.
- (F2) Conservation violation: statistically significant disagreement of $\int_{\Gamma} \omega$ between homotopic paths within a verified equilibrium region.
- (F3) Hamiltonian coherence: volume-preserving dynamics exhibiting the same coherence–exploration coupling as dissipative systems.
- (F4) Resolution without care: contradiction resolution along paths confined to low-/zero-care regions (as operationalized).

(F5) Universal uniqueness: systematic absence of multiplicity where the heuristic non-uniqueness picture predicts multiple coherent configurations.

Remark 5.4 (Protocols). Concrete testing protocols for large language models and other high-dimensional systems can be specified by choosing: (i) a coherence observable φ (e.g. compressibility/entropy/intrinsic dimension proxies), (ii) a feasible estimator for $\log |\det \nabla F|$ (e.g. factorization across steps, trace estimators, or low-rank sketches), and (iii) statistical thresholds for correlation and robustness checks.

6 Discussion and limitations

Mathematical contributions

The core mathematical results are:

1. **Existence (Theorem 3.11):** The autonomous viability model with saturation (2) admits weak coherent configurations under mild coefficient assumptions.
2. **Nontriviality (Theorem 3.15):** When the viability threshold is exceeded ($\lambda_1(-\Delta - b; M) < 0$), there exists a positive solution. This gives a precise criterion for when coherent presence can be self-maintained.

Mathematical limitations

The saturation term $-c\Phi^p$ is necessary to break positive homogeneity and enable nontrivial Dirichlet solutions. Without it (the V1 limiting case), the maximum principle forces only the trivial solution. This is a structural constraint, not a proof gap.

Conceptual limitations

The pointwise CD constraint and the PDE proxy are not proven equivalent; mapping between them is an open problem. The canonical viability closure $b = \kappa\gamma - \lambda\mu$ is a modeling choice; other closures may be appropriate in different contexts.

Operational limitations

The CD condition and falsifiability protocols require careful pre-registration of observables and estimators to avoid tautological fitting; they are empirical proposals.

Relationship to existing frameworks

The CD framework shares conceptual DNA with several existing formalizations of self-organization:

- **Autopoiesis** [13]: CD’s “viability threshold” operationalizes the autopoietic notion of self-maintenance through boundary conditions.
- **Free Energy Principle** [14]: Both frameworks formalize how systems maintain coherence against environmental perturbation; CD emphasizes geometric/topological structure while FEP emphasizes information-theoretic bounds.
- **Enactivism** [15, 16]: The “semiotic manifold” reflects enactivism’s emphasis on meaning as emergent from organism-environment coupling.

- **Dissipative structures** [17]: The viability threshold condition is analogous to criticality conditions for far-from-equilibrium self-organization.

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